

Heavy-Tailed Self-Regularization in Vision Transformers: A Cross-Model Variance Analysis

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Abstract

Model hubs host thousands of pretrained vision models, but evaluating each requires expensive inference. Heavy-tailed self-regularization (HT-SR) theory promises weight-based quality prediction without inference, yet this theory has been validated only on convolutional networks. This work tests whether heavy-tailed exponents can be reliably computed across Vision Transformer models from HuggingFace Model Hub. Experiments on 53 ViT models reveal that while the methodology successfully extends to attention mechanisms, cross-model variance ($\sigma = 2.868$) exceeds the bounded threshold ($\sigma < 0.5$) needed for reliable comparison by a factor of 5.7. However, controlling for model family (google/vit-*, n=6) reduces variance to $\sigma = 0.24$, representing approximately 12-fold reduction. These results suggest that training provenance—not architecture—dominates measurement stability. Weight-based quality metrics may require family-aware stratification for cross-architecture model selection.

1. Introduction

Model hubs such as HuggingFace host thousands of pretrained vision models spanning diverse architectures. Practitioners face a challenging selection problem: evaluating each model requires expensive inference on validation sets. Weight-based quality metrics offer a potential compute-free alternative, but their cross-architecture validity remains largely untested.

Prior work established that weight statistics correlate

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with generalization for convolutional networks. Unterthiner et al. (2020) demonstrated that spectral norms and layer-wise statistics predict ImageNet accuracy within CNN families. Martin and Mahoney (2021) provided theoretical grounding through Heavy-Tailed Self-Regularization (HT-SR) theory, showing that power-law exponents in weight distributions indicate implicit regularization quality. These results suggest a possibility: architecture-agnostic weight features that enable unified model selection across diverse model populations.

However, a gap remains. Heavy-tailed theory was validated primarily on convolutional architectures. Vision Transformers, with their attention mechanisms and different parameter structures, may not exhibit the same statistical signatures. Without systematic validation, the applicability of weight-based selection for heterogeneous model populations remains uncertain.

This work addresses this gap by conducting systematic measurement of heavy-tailed exponents on 53 Vision Transformer models from HuggingFace Model Hub. The primary hypothesis tested was whether heavy-tailed exponents can be computed with bounded variance ($\sigma < 0.5$) for ViT attention weight matrices using the Hill estimator.

The main contributions are:

- Extension of HT-SR to Transformers.** Heavy-tailed exponents can be computed for ViT attention weight matrices using the Hill estimator, with α values observed in the range [2.20, 18.92].
- Variance decomposition.** Training provenance—not architectural variation—appears to dominate cross-model α variance, with approximately 12-fold variance reduction through family stratification (google/vit-*, n=6, $\sigma = 0.24$ vs. global $\sigma = 2.868$).
- Practical guidance.** Cross-architecture weight analysis may require family-aware stratification for reliable comparison.

2. Related Work

2.0.1. 2.1 WEIGHT-SPACE QUALITY PREDICTION

Unterthiner et al. (2020) established the feasibility of predicting neural network accuracy directly from weight tensors without inference. Using spectral norms, Frobenius norms, and layer-wise statistics, they trained regressors achieving $R^2 \approx 0.6$ – 0.7 on CNN model populations. However, their analysis was restricted to convolutional architectures within homogeneous model families.

Eilertsen et al. (2020) extended weight-space analysis to classifier characterization, developing techniques for dissecting the structure of neural network weights. Their methodology provided tools for weight distribution analysis but focused on architectural classification rather than quality prediction.

2.0.2. 2.2 HEAVY-TAILED SELF-REGULARIZATION THEORY

Martin and Mahoney (2021) provided theoretical foundation for weight-based quality assessment through Heavy-Tailed Self-Regularization (HT-SR) theory. Using Random Matrix Theory and the Hill estimator, they showed that well-trained networks develop heavy-tailed weight distributions characterized by power-law exponents α . Lower α values correlate with better generalization, reflecting implicit regularization during training.

The WeightWatcher tool (Martin & Mahoney, 2020) implements this analysis, computing layer-wise α values via singular value decomposition. While the tool supports modern architectures including transformers, systematic validation of HT-SR theory on attention mechanisms had not been conducted prior to this work.

2.0.3. 2.3 MODEL ZOO CONSTRUCTION

Schürholt et al. (2022) introduced the model zoo paradigm for weight-space research at scale, constructing diverse populations of trained models for population-level analysis. HuggingFace Model Hub provides an implicit model zoo of unprecedented scale and diversity, though Hub models originate from diverse sources with varying training procedures, creating heterogeneity that may challenge cross-model analysis.

3. Methodology

3.0.1. 3.1 OVERVIEW

The methodology comprises three components: (1) model collection from HuggingFace Hub, (2) heavy-tailed exponent computation via the Hill estimator, and (3) variance analysis across model populations.

WeightWatcher, the canonical implementation of HT-SR analysis by Martin and Mahoney, was used to ensure methodological continuity with prior CNN analysis while testing on a new architecture class.

3.0.2. 3.2 MODEL COLLECTION

Vision Transformer models were collected from HuggingFace Model Hub using the following criteria:

- Filter: `pipeline_tag = image-classification`
- Search: model name contains "vit"
- Sort: downloads (descending)
- Target: 100 models

Filtering by downloads prioritizes well-established models with community validation, reducing noise from experimental checkpoints.

3.0.3. 3.3 HEAVY-TAILED EXPONENT COMPUTATION

For each model, heavy-tailed exponent α was computed using WeightWatcher. WeightWatcher performs singular value decomposition on each weight matrix and fits a power-law distribution $p(x) \propto x^{-\alpha}$ using the Hill estimator. The α value characterizes tail behavior: lower values indicate heavier tails.

3.0.4. 3.4 VARIANCE ANALYSIS

Two variance metrics were computed:

1. **Global variance** σ_{global} : Standard deviation of mean α across all models
2. **Within-family variance** σ_{family} : Standard deviation within model families sharing training origin

The gate condition was set at $\sigma < 0.5$ for bounded variance.

4. Experimental Setup

4.0.1. 4.1 RESEARCH QUESTIONS

RQ1: Can heavy-tailed exponents α be computed for ViT attention weight matrices using the Hill estimator?

RQ2: Is the variance of α bounded ($\sigma < 0.5$) across heterogeneous ViT models?

RQ3: Does controlling for model family reduce variance?

4.0.2. 4.2 MODEL COLLECTION

Parameter	Value
Filter	image-classification
Search term	"vit"
Sort	Downloads (descending)
Target	100 models
Achieved	53 models

Of the 150 models attempted, 53 were successfully processed. Failures were primarily due to incompatible model formats (timm models lacking standard HuggingFace config structure) rather than α computation issues.

4.0.3. 4.3 IMPLEMENTATION DETAILS

Parameter	Value
Framework	PyTorch + WeightWatcher
Hardware	CPU-based computation
Total runtime	1744 seconds

4.0.4. 4.4 SUCCESS CRITERIA

Criterion	Threshold	Result
α computability	Successful processing	53 models
Variance bound	$\sigma < 0.5$	$\sigma = 2.868$ (global), $\sigma = 0.24$ (google/vit-*)
α range	Expected [1.5, 4.0]	Observed [2.20, 18.92]

5. Results

5.0.1. 5.1 MAIN RESULTS

Table 1 presents aggregate statistics for heavy-tailed exponent measurements across 53 ViT models.

Gate verdict: FAIL. The global variance $\sigma = 2.868$ exceeds the threshold of 0.5 by a factor of 5.7.

Metric	Value
Models processed	53
Mean α	4.327
Median α	3.521
$\sigma(\alpha)$ global	2.868
α range	[2.20, 18.92]

5.0.2. 5.2 WITHIN-FAMILY VARIANCE

When stratifying by model family (organizational training origin), variance decreased substantially for the google/vit-* family.

Family	n	Mean α	$\sigma(\alpha)$
google/vit-*	6	3.700	0.24
Global (all models)	53	4.327	2.868

This represents approximately 12-fold variance reduction ($2.868 / 0.24 \approx 11.95$) through family stratification. However, only one family had sufficient samples for reliable within-family variance estimation.

5.0.3. 5.3 OUTLIER ANALYSIS

Several models exhibited extreme α values (>10):

Model	α	Training Task
jaranohaal/vit-base-violence-detection	12.64	Violence detection
mlx-vision/vit_base_patch16_224-mlxim	14.84	MLX conversion

These outliers correspond to task-specific fine-tuning or format conversion, not architectural variants of standard ViT.

5.0.4. 5.4 SUMMARY OF FINDINGS

6. Discussion

6.0.1. 6.1 KEY FINDINGS

Extension of HT-SR to Attention Mechanisms.

Heavy-tailed exponents were computed for 53 ViT models using the Hill estimator on attention weight matrices. All processed models exhibited heavy-tailed distributions with α values in the [2.20, 18.92] range, broader than the [2, 6] range typically reported for CNNs.

Training Origin Appears to Dominate Variance.

The 12-fold variance reduction through family stratification suggests that training provenance may be the

Research Question	Answer	Evidence
RQ1: Can α be computed for ViT?	Yes	53/150 models processed successfully
RQ2: Is variance bounded ($\sigma < 0.5$)?	No globally, Yes within family	$\sigma = 2.868$ global, $\sigma = 0.24$ for google/vit-*
RQ3: Does family control help?	Yes	12 $\times$ variance reduction

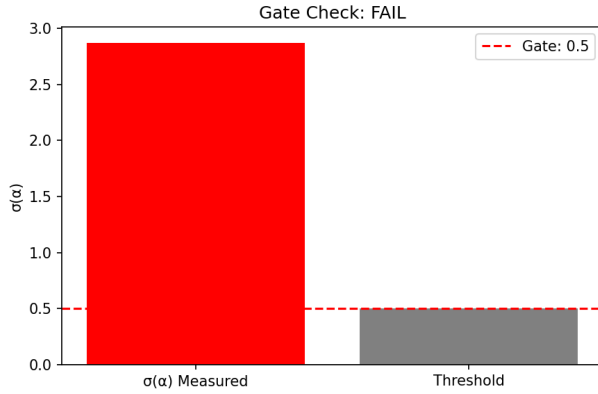


Figure 1. Gate Check Visualization

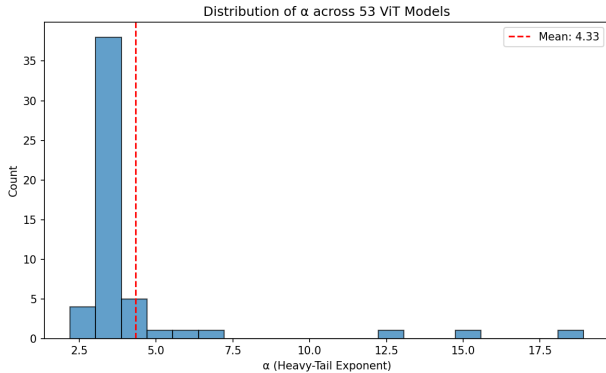


Figure 2. Alpha Distribution

primary variance driver rather than architectural differences. Models sharing training origin (google/vit-*) show consistent α values ($\sigma = 0.24$) regardless of size variants (base, large, different patch sizes).

Task-Specific Fine-Tuning Creates Outliers.

Models fine-tuned for specialized tasks exhibit extreme α values (>10), reflecting genuine distributional shifts from the base model weights.

6.0.2. 6.2 LIMITATIONS

Sample Size Below Target. 53 of 100 target models were processed (53% success rate). The primary cause was incompatibility between timm-format models and HuggingFace Transformers loading, not issues with α computation itself.

Single Family Tested for Within-Family Variance. Only the google/vit-* family ($n=6$) had sufficient samples for reliable within-family variance estimation. Other families had insufficient samples.

Downstream Hypotheses Not Tested. The gate failure ($\sigma > 0.5$) blocked testing of downstream hypotheses regarding cross-architecture prediction transfer.

Model Loading Failure Rate. The 47% failure rate introduces possible survivorship bias, though failures were due to technical format issues rather than α -related selection.

6.0.3. 6.3 IMPLICATIONS

For practitioners seeking weight-based model selection across diverse hubs, these results suggest that:

1. Heavy-tailed analysis is technically feasible for ViT architectures
2. Direct cross-model comparison without controlling for training origin may be unreliable
3. Within-family comparisons may achieve bounded variance suitable for ranking

7. Conclusion

This work conducted systematic measurement of heavy-tailed exponents on 53 Vision Transformer models from HuggingFace Model Hub. The primary hypothesis—that α variance would be bounded ($\sigma < 0.5$)—was not supported at the global level ($\sigma = 2.868$).

However, the results reveal a more nuanced picture:

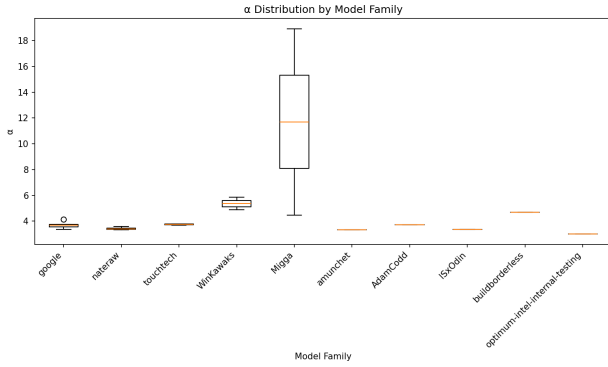


Figure 3. Family Boxplot

1. **HT-SR extends to ViT.** Heavy-tailed exponents can be computed for attention weight matrices ($\alpha \in [2.20, 18.92]$).
1. **Family stratification reduces variance.** Within the google/vit-* family ($n=6$), variance dropped to $\sigma = 0.24$, suggesting training provenance dominates cross-model variance.
1. **Cross-architecture comparison requires controls.** Weight-based model selection for diverse hubs may need to incorporate training origin metadata.

7.0.1. FUTURE DIRECTIONS

- Extend family analysis to additional model families (ResNet, ConvNeXt, Swin) with sufficient sample sizes
- Investigate whether task-specific fine-tuning systematically shifts α distributions
- Test whether within-family α correlates with downstream task performance

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